



Department of Computer Science and Engineering  
Jashore University of Science & Technology  
3<sup>rd</sup> Year 1<sup>st</sup> Semester Final Examination-2024  
Session: 2020-21

Tamim Sheikh  
ID-200119

Course Code: CSE 3103

Time: 3.00 hours

Course Title: Computer Networks and IoT

Full marks: 72

[Figures in the right margin indicate full marks. Answer any SIX questions]

1. a) Define computer network. Why is the computer network so important for modern life? 3  
b) "A star topology is less expensive than a mesh topology." - Is the statement true? Justify your answer. 3  
c) Define protocol. Briefly describe functions of each layer in the OSI model. 6
2. a) Define network throughput and bandwidth. Write advantages and disadvantages of optical fiber transmission 5  
b) Describe different types of data transmission modes. 5  
c) What is the limitation of a hub over a switch? 2
3. a) Explain Logical address, Physical address, and Port address with examples. 3  
b) Given the dataword 1010011010 and the divisor 10111. 6  
i) Show the generation of the CRC codeword at the sender site (using binary division).  
ii) Show the checking of the CRC codeword at the receiver site (assume no error)  
c) Describe different persistent methods in CSMA protocol. 3
4. a) What is IP address? Why do we need IPv6 over IPv4? 2  
b) Suppose a block of addresses is granted to a small organization and one of the addresses is 205.16.37.39/28. 10  
i) What is the first address in the block?  
ii) What is the first usable address in the block?  
iii) What is the last usable address in the block?  
iv) What is the broadcast address in the block?  
v) Find the number of addresses in the block.
5. a) Explain Classful and Classless IP addresses with their pros and cons. 6  
b) Break up 192.168.1.64/28 to support as many subsets as possible with 8 hosts per subnet. Also, show the network address, subnet mask, first usable address, last usable address, broadcast address, and how many addresses are supported in each block. 6
6. a) "Internet is a connectionless network." - describe the statement. 3  
b) Describe subnetting and supernetting with examples. 3  
c) Describe the logical to physical address mapping using ARP protocol. 3  
d) What are the Mask, Net-id, and Host-id in computer network? 3

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7. a) How is plain text transformed into ciphertext using encryption algorithms? 2  
b) Explain the structure and key length of DES. What are its strengths and weaknesses? 4  
c) Compare AES with DES in terms of key length, security, and performance. 3  
d) How does a digital signature algorithm work, and what are its components? 3

8. a) Discuss the key components that make up an IoT system and their roles. 2  
b) How does IoT enable the concept of "smart" devices and environments? 3  
c) Explain how IoT contributes to data-driven decision-making and predictive analytics in various fields. 3  
d) Discuss importance of protocol standardization in ensuring interoperability and scalability in IoT networks. 4



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Course Code: CSE 3107  
Course Title: Software Engineering

Time: 3.00 hours  
Marks: 72

(Note: Answer any six sets of questions from the following eight sets of questions)

1. a) Define heterogeneity in software engineering. What is an SRS? In which stage of software development life cycle, it is required? 4  
b) Differentiate among incremental, iterative, evolutionary, and parallel process flow in software engineering. 8
2. a) What is an SRS? In which stage of software development life cycle, it is required? 2  
b) Write a program in C++/python to detect prime numbers using Sieve method and perform the following: 2+8  
i. Draw the control flow graph of your program.  
ii. Calculate Cyclomatic complexity for your program.  
iii. List all the independent paths in your program.  
iv. Design test cases from each independent path.
3. a) Briefly explain how software requirement analysis is linked with software engineering and software design. 4  
b) Describe the principles of software design. 5  
c) Discuss the differences between verification and validation. 3
4. a) Differentiate between functional and nonfunctional requirements of software. 3  
b) Briefly explain the spiral model. Provide a scenario where spiral model would be preferable in software development. 4  
c) Explain Halstead software metrics and calculate the Halstead metrics (program length, estimated program length, program vocabulary, and purity ratio) for the following code snippet. 5
- ```
def factorial(n):  
    if n == 0:  
        return 1  
    else:  
        return n * factorial(n - 1)
```
5. a) Explain *include* and *extend* keywords in use case diagram with example. 3  
b) Draw a use case diagram for a Library Management System, including key interactions and actors such as librarians, members, and the library database. Specify use cases such as searching for books, borrowing books, returning books, managing member accounts, and any other relevant functionality. 4  
c) Describe different types of coupling and cohesion in software engineering 5

- ~~6.~~ a) Differentiate between white box testing and black box testing. 3
- b) Describe how to test different types of loops in loop testing. 4
- c) Create a class diagram for an Online Banking System. Include key classes such as Customer, Account, Transaction, and Bank. Define the attributes and methods for each class, and show the relationships between the classes, such as associations, inheritances, and dependencies. Ensure to include important functionalities like managing accounts, processing transactions, and generating account statements. 5
- ~~7.~~ a) Briefly explain the MVC framework. Mention at least three popular MVC framework name. 3
- b) Describe agile software development process. 4
- c) Draw a sequence diagram for a Hotel Booking System. Include interactions between key objects such as Customer, Booking System, Payment Gateway, and Hotel Staff. Specify the sequence of messages for booking a room, confirming the booking, processing the payment, and notifying the hotel staff. 5
- ~~8.~~ a) Write a short note on each of the following: 6\*2
- i. Sandwich testing
  - ii. Smoke testing
  - iii. Penetration testing
  - iv. Regression testing
  - v. Alpha testing
  - vi. Integration testing



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ID-200119

Course Code: CSE 3101

Course Title: Operating System and System Programming

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Define interrupt in operating system. Differentiate between interrupt and polling. 03
- b) Explain Linux process tree with diagram. Describe how processes are created and terminated in Linux. 04 10
- c) Define trap in operating system. Explain the steps in making a system call with necessary diagrams. 05
2. a) Define symmetric and asymmetric multiprocessing in operating system. Differentiate between thread and process. 03
- b) Write a short note on each of the following Linux command: 04 11
  1. touch
  2. chmod
  3. cat
  4. pwd
- c) Define context switching. Explain all the process states in an operating system along with the process state diagram. 05
3. a) Explain why zombie processes exist and how they are created. Describe how you can prevent the creation of zombie processes. 03
- b) Differentiate between `wait()` and `waitpid()` system calls in Unix. Mention their necessity in terms of Unix kernel based operating systems. 04
- c) Write a C program that creates a child process using `fork()`. The child process should print "Hello from child" and exit, while the parent process should wait for the child to finish and then print "Child has finished". 05
4. a) What is a signal in the Unix system? Mention at least six signals found in most Unix systems. 03
- b) Explain the three possible ways to handle a kernel generated **SIGCHLD** signal by parent process. 04 8
- c) A disk contains 200 tracks (0-199). The request queue contains track no 82, 170, 43, 140, 24, 16, 190 respectively. Current position of R/W head is 50. Calculate the total number of track movement by R/W head using C-SCAN algorithm. 05
5. a) Define deadlock. Differentiate between livelock and starvation. 03
- b) Mention the four necessary conditions for occurring deadlock. Explain the methods for handling deadlock. 04 12
- c) A program has four processes (P0, P1, P2, P3) and four resource types (A, B, C, D). The following table shows the maximum need and currently allocated resources for each process: 05

| Process | Allocation (A, B, C, D) | Max Need (A, B, C, D) |
|---------|-------------------------|-----------------------|
| P0      | 0, 1, 0, 2              | 7, 5, 3, 3            |
| P1      | 2, 0, 0, 0              | 3, 2, 2, 2            |
| P2      | 3, 0, 2, 2              | 9, 0, 2, 2            |
| P3      | 2, 1, 1, 1              | 2, 2, 2, 2            |

The total available resources are {3, 3, 2, 2}. Use the Banker's algorithm to determine if the system is in a safe state or not. If it's in a safe state, show the sequence in which the processes can complete safely.

7, 2, 3, 5  
 3 3 2 2  
 10 5 5 7

6. a) Explain the role of dispatcher and CPU scheduler in operating System. Explain whether they are same with proper reasoning. 03

b) Does the following instruction set create a deadlock? Explain with proper diagram. 04

Process 1 holds R and S.  
Process 2 holds nothing but wants T.  
Process 3 holds nothing but wants S.  
Process 4 holds U and wants S and T.  
Process 5 holds T and wants V.  
Process 6 holds W and wants S.  
Process 7 holds V and wants U

c) Consider a system with the following processes, each with their arrival time (AT), burst time (BT), and quantum = 4. 05

| Process | Arrival Time (AT) | Burst Time (BT) |
|---------|-------------------|-----------------|
| P1      | 0                 | 6               |
| P2      | 1                 | 4               |
| P3      | 2                 | 9               |
| P4      | 3                 | 5               |
| P5      | 4                 | 2               |

Using the Round Robin scheduling algorithm, compute the average waiting time and turnaround time for these processes.

7. a) What do you understand by atomic instructions? Why is it necessary to solve critical section problem? 04

b) Assume you have the following processes to execute with one processor, with the processes arriving at the following times and having the following CPU burst times and priorities: 06

| Process | Arrival Time | CPU Burst Time | Priority |
|---------|--------------|----------------|----------|
| A       | 0            | 8              | 3        |
| B       | 3            | 4              | 1 (H)    |
| C       | 5            | 7              | 4        |
| D       | 8            | 3              | 2        |

For each of the following scheduling algorithms:

- Shortest Job First (SJF), Preemptive
- Priority, Preemptive

1. Draw a Gantt chart. 2. Calculate average waiting time. 3. Calculate turnaround time of each process.

c) Write down the advantage and disadvantages of Round Robin Scheduling algorithm. 02

8. a) "Raid level 01 or 10 increase disk access performance and availability", discuss with a suitable example. 03

b) Determine the total head movement for algorithms SSTF and C-Look for the following requests (in track number): 98, 181, 23, 65, 122, 14, 72, 36, 26, 67; where the current position of head is on track number 0. 06

c) A disk has an average seek time of 5ms, a rotational speed of 15,000 rpm, and 500 sectors per track. What is the average access time to read a single sector? What is the expected time to read 500 contiguous sectors on the same track? What is the expected time to read 500 sectors scattered over the disk? 03



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Tamim Sheikh  
ID-200119  
63

Course Code: CSE 3105

Course Title: Microprocessor & Embedded System

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Define Microprocessor and Microcomputer. 2
- b) Determine the carry, sign, overflow and zero flags for the following operation: 3  

$$\text{ADD } 82\text{AH}, 231\text{FH}$$
- c) Write the comparisons among Intel Core i3, Core i5 and Core i7 microprocessor. 4
- d) Describe the Status Register of the MC68000 microprocessor. 3
2. a) Define programed I/O and I/O programed. 2
- b) Determine the effect of each one of the following 8086 instructions: 5
  - i. PUSH [BX]
  - ii. DIV DH
  - iii. CWD
  - iv. MOVS
  - v. MOV START [BX], AL

Assume the following data prior to execution of each of the above instructions independently. Assume all numbers in hexadecimal.

[DS]=3000H, [ES]=5000H, [DX]=0400H, [SP]=5000H, [SS]=6000H, [AX]=00A9H, [SI]=0400H, [DI]=0500H, DF=0, [BX]=6000H,

Value of START=05H.
- c) Write the comparative study between single core and multi core processor. 2
- d) Write an assembly language program for 8086 microprocessors to multiply two 16-bit binary numbers to give a 32-bit result. 3
3. a) What are the features of 8086 microprocessor? 3
- b) Draw and describe the pins and signals specification of 8086 microprocessor. 7 11
- c) What are the tasks of SMC and STI command for 8086 microprocessors? 2
4. a) Draw and describe the internal architecture of 8086 microprocessor. 6 12
- b) Describe the addressing modes of 8086 microprocessor. 6
5. a) Describe with example the Control Transfer Instructions of 8086 microprocessor. 4 11
- b) Describe the status of each field of flag register of 8086 microprocessor. 4
- c) Write the comparisons between 8086 and 8088 microprocessors. 4
6. a) Draw and explain the internal block diagram of 80186 microprocessor. 5 12
- b) What are the salient features of 80286 microprocessor? 3
- c) Describe the status of flag register of 80286 microprocessor. 4
7. a) Briefly explain function of the following. Also give on example each 4 10
  - i) PLD ii) COTS
- b) Define memory map? Explain the interrupt map for embedded system. 4
- c) Design metrics for a typical embedded system. 4
8. a) Calculate the effective address for the following register: 4 7
 

SS: 3864H, SP: 1735H, BP: 4826H
- b) Write an assembly language program for comparing two string and concatenating two strings. 4
- c) Write down the advantages of RISC Processor over CISC Processor. 4